

Synopsis: Late-Universe Acceleration from Hysterical Response

Eventual criticality, vacuum attractor, and a one-parameter background test

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Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

This is a short guide to the late-universe cosmological response paper. It assumes the response-theory synopsis. Each section says *why* the claim matters, then states the result. Proofs and the full status table live in the full paper.

1 Why a late-universe continuation

Galactic modelling organizes the residual gravitational response spatially. It does not by itself produce a homogeneous vacuum-like term. The question here is whether dilution on a homogeneous background can force criticality and then a Λ -like leftover.

2 Eventual criticality (homogeneity is not enough)

Paper 1's closed-loop gain is $G_H = \chi_R \chi_H$, critical at $G_H = 1$. On a homogeneous FLRW background with conserved $n \propto a^{-3}$, if the slow gap vanishes as $\gamma = Kn^\alpha$ ($\alpha > 0$) with slowly varying positive residue Γ_* , then

$$G(n) = \frac{\Gamma_*}{Kn^\alpha}$$

is strictly decreasing from ∞ to 0 as n runs from 0 to ∞ .

Theorem 2.1 (Eventual dilution criticality). *There is a unique $n_c = (\Gamma_*/K)^{1/\alpha}$ and a unique a_c such that the expanding universe is subcritical for $a < a_c$ and supercritical for $a > a_c$.*

Homogeneity supplies a single $n(a)$; the gap law forces the crossing. A density-independent gap need never become critical. For $\alpha = 1$ one recovers the modelling law $G = (a/a_c)^3$ used below.

3 Finite-amplitude crossing

Once linearly unstable, enough integrated growth forces every intermediate amplitude to be visited.

Theorem 3.1 (Cosmological finite-amplitude crossing). *If $\dot{X} = \lambda(t)X$ and $\int \lambda \geq \ln(X_*/X_i)$ with $X_* > X_i > 0$, then some t_* has $X(t_*) = X_*$.*

4 Nonlinear attractor

A cubic pitchfork has stable branches $\phi_*^2 = (r_0/u)(G-1)$ for $G > 1$. With the modelling closure $\rho_H = C\rho_m\phi^2$ and $\rho_m \propto a^{-3}$,

$$\rho_H^*(a) = A(a_c^{-3} - a^{-3}) \longrightarrow Aa_c^{-3}, \quad w_H = -1 - \frac{1}{G-1} \rightarrow -1$$

from below. Present density ratios calibrate only Cr_0/u once z_c is chosen.

5 Covariant positivity and slow-pole closure

The same attractor in density variables gives $\rho_H = A(n_c - n)$ and, under a minimal fluid action, $p_H = -\rho_{H,\infty}$. A slow-pole closure with reciprocal projection sharpens this to $\rho_{H,\infty} = mn_c$ and local slope $B = 3$.

6 Background confrontation

Fixing today's Ω_{m0}, Ω_{H0} to a Planck 2018 flat baseline, the attractor has one free shape parameter z_c . Late criticality ($z_c \sim 1$) produces distance-modulus residuals of several hundredths of a magnitude versus Λ CDM; $z_c \gtrsim 3$ –5 approaches Λ CDM at the millimagnitude level. That is a one-parameter shape test, not a derivation of Ω_{H0} from K and Γ_* .

7 Gravity versus electromagnetism

Homogeneous gravitational mass-energy need not neutralize. Exact electric-charge conservation forbids generating a nonzero homogeneous net charge from $Q = 0$.

8 What is not claimed

No microscopic Standard-Model evaluation of K, Γ_* ; no full SN/BAO/CMB likelihood; no joint galactic/Solar/cosmological posterior; no claim that observed acceleration has been shown to be Hysterical Response.